

Olga Isupova

Address	Apt 180, 153 Great Ancoats Street, Manchester, M4 6DT	Mobile Phone	+44 7922 737699
Date of Birth	1 st December 1989	Email	ihoho89@gmail.com
Place of Birth	Krivoy Rog, USSR		
Nationality	Russian		

Education

- 2007--** *Specialist in Mathematics and System Programming* with Honours, degree equals to Masters
2012 [Lomonosov Moscow State University, Russia](#)
Faculty of [Computational Mathematics and Cybernetics](#), Department of [Mathematical Methods of Forecasting](#)
Thesis: *Hierarchical topic modeling in document collections by probabilistic latent semantic analysis*.
Thesis is devoted to the building a topic hierarchy tree from unlabelled document collection.
Supervisor: [Dr Konstantin Vorontsov](#), Lecturer at the Lomonosov Moscow State University
- 2010--** Finished [Yandex School of Data Analysis](#), Russia
2013 Program of [Data Analysis](#)
Study on algorithms and data structures, probabilistic and statistical methods in decision making theory, machine learning theory, convex theory and optimisation, linguistics and machine translation and others
- 2014--** *PhD*
Present [The University of Sheffield](#), UK
Department of [Automatic Control and Systems Engineering](#).
Thesis: *Abnormal behaviour detection in video*. Developing methods for autonomous video scene interpretation and abnormal event detection. Different methods and techniques are studied and applied, including topic modeling, Bayesian nonparametrics, change point detection, approximated and sampling inference scheme, image and video processing.
Supervisor: [Dr. Lyudmila Mihaylova](#), Reader in Signal Processing and Control, Associate Professor at the University of Sheffield.

Employment History

- 04.2014 -** *Marie Curie Early Stage Researcher at* [Marie Curie FP7 ITN TRAX project](#)
Present [The University of Sheffield](#)
Research topic is ``Knowledge Extraction from data''. Development of new algorithms for information extraction from complex, heterogeneous data. Design of a new theory providing computationally feasible ways to uncover the underlying structures in the data.
- 03.2011 -** *Research Associate*
02.2014 [Croc Inc](#)
Machine learning and data mining research and development: design of data mining algorithms, outliers filtering algorithms; data pre-processing.
Business application functional implementation: design of business processes for data analysis, automation of developed algorithms.
- 06.2011 -** *Analyst of the Customer Relationship Management department*
09.2011 [Tehnosila](#)
Data analysis in retail operations: sampling customers for target marketing actions; analysis of communication channels, results of marketing actions; statistical and econometric analysis of customers' database.

Publications

Journals

O. Isupova, D. Kuzin, L. Mihaylova. "*Learning Methods for Dynamic Topic Modeling in Automated Behaviour Analysis*", in IEEE Transactions on Neural Networks and Learning Systems, **under review**

Conferences

Z.Li, **O. Isupova**, L. Mihaylova, L.Rossi. "*Autonomous Flame Detection in Video Based on Saliency Analysis and Optical Flow*", in the Proceedings of the IEEE International Conference on Multisensor Fusion and Integration for Intelligent Systems, 19-21 September 2016, Baden-Baden, Germany

O. Isupova, D. Kuzin, L. Mihaylova. "*Dynamic Hierarchical Dirichlet Process for Abnormal Behaviour Detection in Video*", in the Proceedings of the 19th International Conference on Information Fusion, 5-8 July 2016, Heidelberg, Germany

D. Kuzin, **O. Isupova**, L. Mihaylova. "*Compressive Sensing Approaches for Autonomous Object Detection in Video Sequences*", in Proceedings of 10th Workshop Sensor Data Fusion: Trends, Solutions, and Applications, 6-8 October 2015, Bonn, Germany

O. Isupova, L. Mihaylova, D. Kuzin, G. Markarian, F. Septier. "*An Expectation Maximisation Algorithm for Behaviour Analysis in Video*", in Proceedings of 18th International Conference on Information Fusion (Fusion), 6-9 July 2015, Washington, USA

O. Isupova, D. Kuzin, L. Mihaylova. "*Abnormal behaviour detection in video using topic modeling*", in Proceedings of University of Sheffield Engineering Symposium, 24 June 2015, Sheffield, UK. Abstract

O. Isupova. "*Matching a Hierarchical Topic Model with External Subject Authorities*", in Proceedings of the abstracts of the XIX International Conference of Students and Young Scientists "Lomonosov-2012" April 2012, Moscow, Russia

Workshops

O. Isupova, D. Kuzin, L. Mihaylova. "*Anomaly Detection in Video with Bayesian nonparametrics*", ICML 2016 Anomaly Detection Workshop, 24 June 2016, New York, USA

Expertise

■ Maths

Probability theory, Statistics, Stochastic processes
Machine learning, Bayesian statistics, Bayesian nonparametrics
Graphical models, Approximate Inference, Sparse models
Mathematical analysis, Convex Optimisation

■ Programming and Miscellaneous

Matlab, R, Python
C/C++
Databases, SQL

Other research related activity

■ Talks

Talk "Modern Video Analytics Systems" at the Sheffield Festival of Science and Engineering as part of British Science week 2016, March 2016;

Talk "Abnormal Behaviour Detection in Video" at the PGR symposium at the University of Sheffield, November 2014

■ Served as chairs and organisers

A session chair at the 19th International Conference on Information Fusion, July 2016;
The organising committee of the summer school for the TRAX project, Sheffield, June-July 2015;
Athena Swan Committee at the department of Automatic Control and Systems Engineering

■ Served as reviewers

Signal Processing Elsevier journal, 2015 American Control Conference, 29th Conference on Neural Information Processing Systems (NIPS 2016), IEEE International Conference on Multisensor Fusion and Integration for Intelligent Systems (MFI 2016)